Fiscal Year:	FY 2008	**Task Last Updated:**	FY 04/01/2009
PI Name:	Landau, David Paul Ph.D.		
Project Title:	Computer Simulations of Confined Quantum Systems in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30602	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2004	**End Date:**	12/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2007 per PI's institution and GRC document (10/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	GSN 4515		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	The objective is to use computer simulations of appropriate models to improve our understanding of the behavior of quantum systems in confined geometries as the phase transition is approached. Our immediate objective is to use the XY-model of "spins" to describe the critical behavior of superfluid helium. This is accomplished by an appropriate renaming of variables that yields a physically relevant description of the lambda point. We are developing improved methods for carrying out Monte Carlo simulations for systems with two parallel surfaces that play the role of the confining walls. Then, new spin dynamics methods will be used to study the time dependent behavior of the model and thus the thermal conductivity. Data will be compared with those from the BEST and DYNAMX experiments. Path integral Monte Carlo methods are also to be developed to study the behavior of confined Boson systems in different types of fields. Both the techniques developed and the results appropriate to helium may play an important role in the elucidation of behavior of other confined quantum systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Path integral Quantum Monte Carlo simulations have been completed for systems in low dimension and for Bosons with dipole moments (e.g. for Cr52). We find that the superfluid fraction for a quasi-2D Bose gas is distinctly different from that for both a quasi-1D and a true 3D system, i.e. the superfluid fraction for a quasi-2D system decreases faster than for the !D case and a true 3D system with increasing temperature and shows a stronger dependence on interaction strength. The superfluid fraction for a quasi-2D gas lie well below those calculated from a semi-classical approximation. Simulations have also been completed for a pseudo-one-dimensional system in a box trap that is similar to a novel trap used in recent experiments. We found that the net effect of the dipole interactions depends upon the trapping geometry. Classical simulations have been completed for xy-systems in quasi-2D geometry that mimics the configuration used for experiments performed to measure the thermal resistivity of 4He. Surprisingly good agreement with experiment was found.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Articles in Peer-reviewed Journals	Nho K, Landau DP. "Finite-temperature properties of quasi-two-dimensional Bose-Einstein condensates." Physical Review A. 2006 Mar;73(3):033606. http://dx.doi.org/10.1103/PhysRevA.73.033606 , Mar-2006
Articles in Peer-reviewed Journals	Zhang C, Nho K, Landau DP. "Finite-size effects on the thermal resistivity of 4He in the quasi-two-dimensional geometry." Physical Review A. 2006 May;73(17):174508. http://dx.doi.org/10.1103/PhysRevB.73.174508 , May-2006
Articles in Peer-reviewed Journals	Nho K, Landau DP. "Finite-temperature properties of binary mixtures of two Bose-Einstein condensates." Physical Review A. 2007 Nov;76(5):053610. http://dx.doi.org/10.1103/PhysRevA.76.053610 , Nov-2007
Articles in Peer-reviewed Journals	Zhang C, Nho K, Landau DP. "Properties of Bose-Einstein condensates in a quasi-one-dimensional box trap." Physical Review A. 2008 Feb;77(2):025601. http://dx.doi.org/10.1103/PhysRevA.77.025601 , Feb-2008
Articles in Peer-reviewed Journals	Nho K, Landau DP. "Superfluid density estimators for a molecular para-hydrogen cluster." submitted, January 2009. , Jan-2009